5.1 Dilations

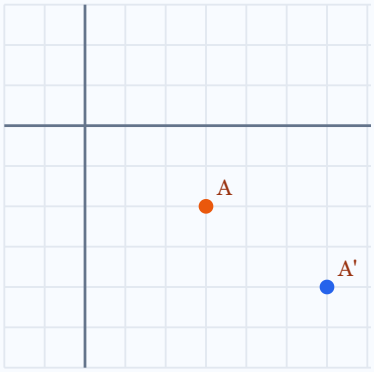
A **dilation** is a transformation that scales a figure from a fixed **center** by a **scale factor** k, producing a similar image. For a dilation centered at the origin, the rule is $(x, y) \mapsto (kx, ky)$; for a center at (a, b), translate the center to the origin first: $(x, y) \mapsto (k(x-a)+a,\ k(y-b)+b)$. When $|k| > 1$ the image is an **enlargement**; when 0<|k|<1 it is a **reduction**; a negative k places the image on the opposite side of the center, along the same line through the center and the original point. Lengths scale by a factor of $|k|$, but **areas scale by** k^2 (since both dimensions grow). The center itself is the only fixed point — it always maps to itself. This worksheet practises dilating points and figures, finding scale factors, and reasoning about how length and area change.

Examples

Example 1: Dilate from the origin

Find the image of $A(3, -2)$ under a dilation centered at the origin with $k = 2$. Classify as an enlargement or reduction.

Solution



A' is twice as far from the origin as A, along the same ray.

Identify the rule for an origin-centered dilation: $(x, y) \mapsto (kx, ky)$.

Apply with $k = 2$: $A' = (2 \cdot 3,\ 2 \cdot (-2)) = (6, -4)$.

Since |k|=2>1, the image is larger than the original.

Answer: $A'(6, -4)$ — an enlargement.

Example 2: Dilate a triangle (reduction)

$\triangle ABC$ has vertices $A(2,4)$, $B(6,4)$, $C(6,-2)$. Find the image under a dilation about the origin with $k=\dfrac{1}{2}$, and classify it.

Solution

Apply $(x,y)\mapsto(\frac{1}{2}x,\frac{1}{2}y)$ to each vertex:

$A'=(1,2),\quad B'=(3,2),\quad C'=(3,-1).$

Since 0<k=\dfrac{1}{2}<1, the image is smaller than the original.

Answer: $A'(1,2), B'(3,2), C'(3,-1)$ — a reduction.

Example 3: Find the scale factor

Under a dilation centered at the origin, $P(4,6)$ maps to $P'(10,15)$. Find k, and verify using both coordinates.

Solution

Since the dilation is centered at the origin, $k=\dfrac{x'}{x}=\dfrac{y'}{y}$.

Using the x-coordinates: $k=\dfrac{10}{4}=\dfrac{5}{2}$.

Check with the y-coordinates: $k=\dfrac{15}{6}=\dfrac{5}{2}$ ✓ — consistent.

Answer: $k=\dfrac{5}{2}$.

Example 4: Dilation about a non-origin center

Dilate $Q(5,3)$ about center $P(1,1)$ with $k=3$.

Solution

Translate so the center becomes the origin: subtract $(1,1)$ from Q: $(5-1,3-1)=(4,2)$.

Scale by $k=3$: $(3\cdot 4,\ 3\cdot 2)=(12,6)$.

Translate back: add $(1,1)$: $(12+1,\ 6+1)=(13,7)$.

Answer: $Q'(13,7)$.

Example 5: Negative scale factor

Dilate $R(3, 5)$ about the origin with $k = -2$. Describe the image's position relative to the center.

Solution

Apply $(x, y) \mapsto (-2x, -2y)$: $R' = (-2 \cdot 3,\ -2 \cdot 5) = (-6, -10)$.

$R(3, 5)$ lies northeast of the origin; $R'(-6, -10)$ lies southwest of the origin — the opposite side, along the same line through the center.

Distance check: R is $\sqrt{34}$ from the origin; R' is $\sqrt{136} = 2\sqrt{34}$, exactly $|k| = 2$ times as far.

Answer: $R'(-6, -10)$, on the opposite side of the center, twice as far away.

Example 6: Length scales by $|k|$

A segment of length 8 is dilated with $k = -1.5$. Find the image length.

Solution

Length always scales by the absolute value of the scale factor, regardless of sign.

Image length $= |k| \times 8 = 1.5 \times 8 = 12$.

Answer: 12.

Example 7: Area scales by k^{2}

A square has side length 4 (area 16). It is dilated with $k = 3$. Find the new side length, new area, and the ratio of the new area to the original. Compare that ratio to k^2.

Solution

New side length $= 4 \times 3 = 12$.

New area $= 12^2 = 144$.

Ratio of areas $= \dfrac{144}{16} = 9$.

Compare: $k^2 = 3^2 = 9$ — matches exactly.

Conclusion: area scales by k^2, not k, because both dimensions of the square are multiplied by k.

Example 8: The fixed point

A dilation is centered at $C(-2, 5)$. What is the image of C itself, no matter what k is?

Solution

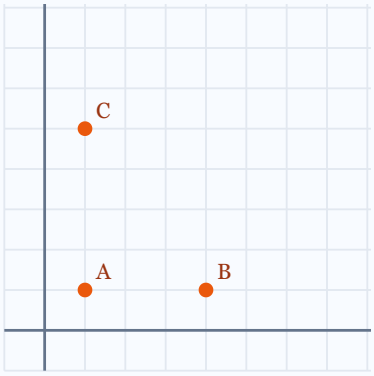
By definition, the center of a dilation is the one point that never moves — every other point scales relative to it, but the center's distance to itself is always 0, and $0 \times k = 0$.

Answer: C maps to itself: $(-2, 5)$.

Example 9: Dilation preserves direction (parallel sides)

Triangle has vertices $A(1, 1)$, $B(4, 1)$, $C(1, 5)$. It is dilated about the origin with $k = 2$. Verify that $\overline{B'C'}$ is parallel to $\overline{BC}$ by comparing slopes.

Solution



Dilating about the origin with $k = 2$ doubles every coordinate.

Find the image points: $A'(2, 2)$, $B'(8, 2)$, $C'(2, 10)$.

Slope of $\overline{BC}$: $\dfrac{5-1}{1-4} = \dfrac{4}{-3} = -\dfrac{4}{3}$.

Slope of $\overline{B'C'}$: $\dfrac{10-2}{2-8} = \dfrac{8}{-6} = -\dfrac{4}{3}$.

The slopes match exactly.

Conclusion: $\overline{B'C'} \parallel \overline{BC}$ — dilations preserve the direction of every side, which is why the image is always similar to the original.

Example 10: Blueprint scale factor

An architect's blueprint uses scale factor $k = \frac{1}{50}$ (from the real building to the drawing). A wall is 200 units long on the actual building. Find its length on the blueprint, and find the scale factor to convert blueprint measurements back to real ones.

Solution

Blueprint length $= k \times (\text{real length}) = \frac{1}{50} \times 200 = 4$ units.

The reverse dilation (blueprint $\rightarrow$ real) uses the reciprocal scale factor: $\frac{1}{1/50} = 50$.

Check: $4 \times 50 = 200$ ✓, back to the original length.

Answer: Blueprint length $= 4$ units; reverse scale factor $= 50$.

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Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: Dilate from origin

Find the image of $A(2, 5)$ under a dilation about the origin with $k = 4$.

Answer: **Verification:**

Problem 2: Dilate from origin

Find the image of $B(-3, 6)$ under a dilation about the origin with $k = \frac{1}{3}$.

Answer: **Verification:**

Problem 3: Scale factor

Under a dilation about the origin, $P(8, 12)$ maps to $P'(2, 3)$. Find k.

Answer: **Verification:**

Problem 4: Scale factor

Under a dilation about the origin, $Q(5, -4)$ maps to $Q'(15, -12)$. Find k.

Answer: **Verification:**

Problem 5: Classify

Is a dilation with $k = 0.6$ an enlargement or a reduction?

Answer: **Verification:**

Problem 6: Length

A segment of length 10 is dilated with $k = 0.4$. Find the image length.

Answer: **Verification:**

Problem 7: Non-origin center

Dilate $S(2,4)$ about center $P(2,0)$ with $k=2$.

Answer: **Verification:**

Problem 8: Triangle image

$\triangle DEF$ has vertices $D(0,0)$, $E(4,0)$, $F(0,6)$. Dilate about the origin with $k=\frac{1}{2}$.

Answer: **Verification:**

Problem 9: Length

A segment of length 14 is dilated with $k=\frac{3}{2}$. Find the image length.

Answer: **Verification:**

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Problem 10: Negative scale factor

Find the image of $C(7, -1)$ under a dilation about the origin with $k = -2$.

Answer: **Verification:**

Problem 11: Non-origin center

Dilate $T(4, 6)$ about center $O(0, 2)$ with $k = \frac{1}{2}$.

Answer: **Verification:**

Problem 12: Scale factor

Under a dilation about the origin, $M(-2, 8)$ maps to $M'(-5, 20)$. Find k.

Answer: **Verification:**

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Problem 13: Challenge — area scaling

A square has side length 6 (area 36). It is dilated with $k = \frac{2}{3}$. Find the new side length, the new area, and verify the ratio of areas equals k^2.

Answer: **Verification:**

Problem 14: Challenge — negative k, non-origin center

Dilate $W(6, 2)$ about center $C(2, 2)$ with $k = -3$. Find the image, then state where it lies relative to W and C.

Answer: **Verification:**

Problem 15: Challenge — scale model

A model car uses scale factor $k = \frac{1}{12}$ of the real car. The real car is 180 in long with a side-profile area of 2880 in^2. Find the model's length and side-profile area.

Answer: **Verification:**

✓ Answer Key

1. $A'(8, 20)$
2. $B'(-1, 2)$
3. $k = \frac{1}{4}$
4. $k = 3$
5. Reduction
6. 4
7. $S'(2, 8)$
8. $D'(0, 0), E'(2, 0), F'(0, 3)$
9. 21
10. $C'(-14, 2)$
11. $T'(2, 4)$
12. $k = \frac{5}{2}$
13. New side $= 4$; new area $= 16$; ratio $= \frac{4}{9} = k^2$
14. $W'(-10, 2)$; on the opposite side of C from W, three times as far away
15. Model length $= 15$ in; model area $= 20$ in^2

Great work — keep practising!